

SDI Intelligence server: the build plan

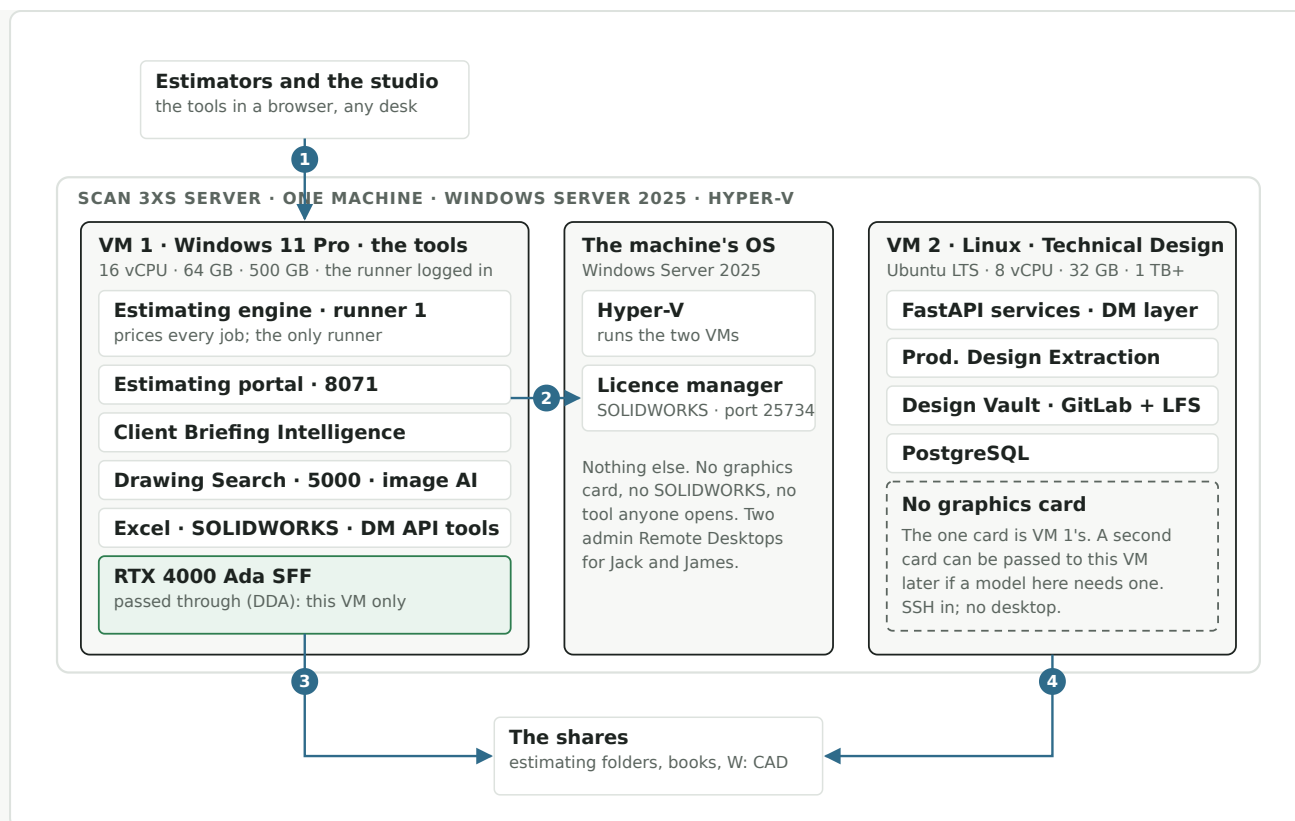
Revised 5 October 2026 · The operating system installed on the machine is WINDOWS SERVER 2025. Windows 11 Pro exists only inside VM 1, the virtual machine that holds the graphics card and the tools people use. One Linux VM for Technical Design. No third VM. Hardware on site Monday 26 October; go-live in the week commencing 2 November.

Jack, SDI's hardware advisor, builds the machine and hands over two virtual machines that behave like new computers on the network: a Windows 11 Pro one with the graphics card inside it, and a Linux one. James, who leads the software design, takes them from there, exactly as he would set up a laptop. The first section says what runs where and why; then Jack's list, the handover checks, James's list, and the questions to settle together.

Scan 3XS · Threadripper PRO 9975WX (32 cores) · 128 GB ECC · NVIDIA RTX 4000 Ada SFF 20 GB · 2 × 2 TB NVMe · Windows Server 2025 host (Hyper-V, SOLIDWORKS licence manager) · VM 1 Windows 11 Pro: the card, estimating, client estimating, Client Briefing, Drawing Search · VM 2 Linux: Technical Design services

What runs where

The operating system on the metal is Windows Server 2025, and it runs nothing a person would use: Hyper-V and the licence manager. The Windows VM is the laptop, moved to the server, with the card: everything that needs a desktop, Excel, SOLIDWORKS or a SOLIDWORKS library. The Linux VM is Technical Design's: GitLab, PostgreSQL and the FastAPI services, which call the Windows VM over HTTPS for anything SOLIDWORKS.



1. People reach every tool on VM 1 in a browser: the estimating portal, Client Briefing and Drawing Search each on its own port. The runner on the same VM takes the oldest job in the queue when it is free.
2. SOLIDWORKS on VM 1 takes a licence from the pool on the host while it is open. The runner keeps it open, so VM 1 holds one seat all day.
3. VM 1 reads and writes the job folders and books on the estimating shares, and reads W: for Drawing Search's index.
4. The Linux VM reads W: for the Document Manager and holds the Design Vault on its own disk.

EACH TOOL, AND WHERE IT LIVES

WHAT	WHERE	WHY THERE
Estimating engine, runner 1, Excel, SOLIDWORKS, the RTX 4000	VM 1 Windows	Excel and SOLIDWORKS are driven over COM, which needs a logged-in desktop; the card is here; SOLIDWORKS does not run on Windows Server.
Estimating portal and queue (port 8071): client estimating	VM 1 Windows	One box for everything estimating, as on the laptop today. The runner points at <code>localhost</code> .
Client Briefing Intelligence	VM 1 Windows	Runs today on a Windows machine; moves as it is.

WHAT	WHERE	WHY THERE
Drawing Search (port 5000): name, brand and image search over the Fixture Library	VM 1 Windows	Indexes W: and opens files on it; the image search uses the card. Its index lives on VM 1's own disk.
Technical Design's Windows parts: the SOLIDWORKS Document Manager API, the CAD Validator, and the COM tools (DXF flat patterns, drawing generation, validation, tolerance checks)	VM 1 Windows	The Document Manager API is a SOLIDWORKS library and runs only on Windows; the COM tools drive SOLIDWORKS itself. The Linux side calls them over HTTPS on one port.
FastAPI services: the Document Manager intelligence layer, Production Design Extraction	VM 2 Linux	Python services with no Windows need; an outage here costs an import, not an estimate.
SDI Design Vault: GitLab with LFS, PostgreSQL, the DM API intelligence layer	VM 2 Linux	GitLab and PostgreSQL are Linux-native; the vault needs its own large disk.
Hyper-V; SOLIDWORKS licence manager	Host	The licence manager is supported on Windows Server and stays up when VM 1 restarts.
The estimating shares and W:	The file server, as now	Nothing here needs this machine to be a file server.

WHY THE HOST IS WINDOWS SERVER AND NOT WINDOWS 11 PRO

Three things were asked of this machine: Remote Desktop into it, the graphics card in a VM, and SOLIDWORKS in a VM. Only a Server host gives all three.

ASKED FOR	WINDOWS SERVER 2025 HOST	WINDOWS 11 PRO HOST
The graphics card used by a VM	Yes. Discrete Device Assignment (DDA) hands the whole card to one VM, with NVIDIA's normal driver inside it.	No. DDA is a Server feature. The card can only drive the host's own desktop; the partitioning Windows 11 has (GPU-P) is unsupported by NVIDIA and Dassault for this card and does not reach Linux guests at all.
SOLIDWORKS in a VM	Yes, in VM 1 with the card. Dassault lists Hyper-V 2025 for SOLIDWORKS 2026.	Only in software graphics.

ASKED FOR	WINDOWS SERVER 2025 HOST	WINDOWS 11 PRO HOST
Remote Desktop	Two administrator sessions on the host at once. The Windows VM allows one desktop, like any Windows 11; the Hyper-V console reaches it without logging the runner off.	One desktop on the host, which is also the runner's.
Can the card be shared, or used by the host and a VM?	No, on either. A card belongs to one operating system at a time. Passed through, it is VM 1's and the host has none, which is fine: a Server host needs no graphics. Sharing one card between VMs needs a data-centre card (NVIDIA L4 and similar) with yearly vGPU licences.	

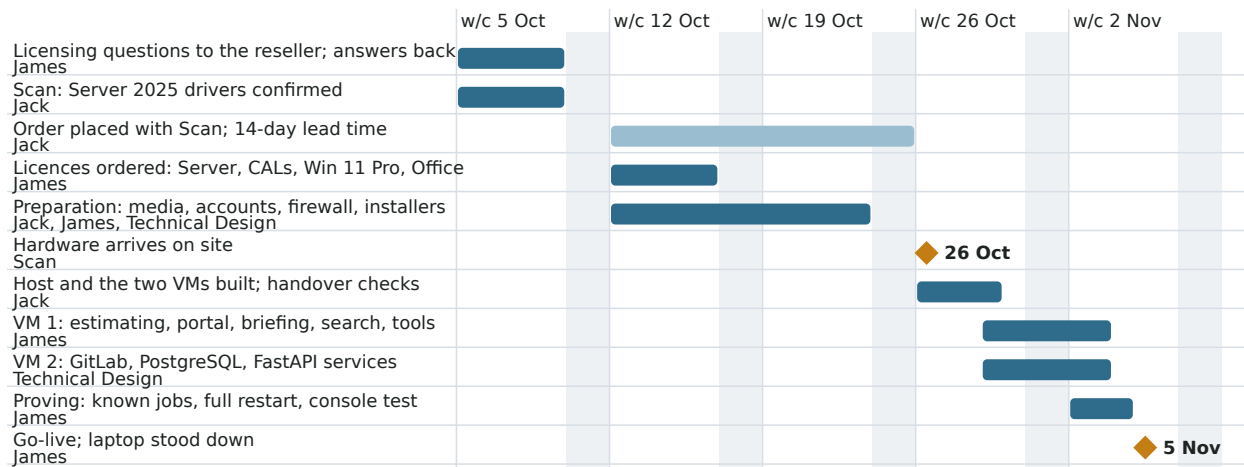
SIZING: WHO GETS WHAT

WHERE	CORES	MEMORY	DISK	NOTE
VM 1 Windows	16 vCPU	64 GB, Dynamic Memory off	500 GB+ on NVMe	SOLIDWORKS, Excel, the engine and Drawing Search's index all here; the models are read here.
VM 2 Linux	8 vCPU	32 GB	1 TB+ on NVMe, growable	The Design Vault's GitLab and LFS store grow with every check-in; size the disk for the whole W: history if the vault is to hold it.
Host	the remaining 8	the remaining 32 GB	the OS drive	Hyper-V and the licence manager need little.

TIMELINE

The order goes to Scan on Monday 12 October, on the reseller's written answer to the first licensing question, for delivery on a 14-day lead time: hardware on site Monday 26

October. Jack builds that week, James and the Technical Design team install the week after, and go-live is in the week commencing 2 November.



Bars run Monday to Friday; diamonds are milestones; shaded columns are weekends. Dates assume written licensing answers by Friday !

WHAT CAN AND CANNOT BE SHARED

The graphics card: one card, one VM.

Passed through (DDA), the RTX 4000 Ada belongs to VM 1 alone; the host and the Linux VM have none. It cannot be split between VMs: Windows Server 2025 only partitions a short list of data-centre cards (NVIDIA L4, L40S, A16 and similar) with NVIDIA's licensed vGPU driver, and this card is not one of them. Moving it to the Linux VM is a shutdown of both VMs each way, so it is for an occasional overnight job, never a routine. If Technical Design ever needs a card of its own, a second card is fitted and passed to the Linux VM the same way; DDA works for Linux guests.

SOLIDWORKS: the licences are shared, the program is not.

SOLIDWORKS runs only on VM 1; the Linux VM reaches its four COM tools over the network. A network (SolidNetWork) licence puts the seats in a pool on the host: SOLIDWORKS takes one when it opens and returns it when it closes, first come, first served, and designers' PCs can share the same pool. A standalone licence works in a VM but is tied to that VM's identity, so a rebuilt VM means a reactivation; the network licence avoids that.

One runner, one estimate at a time.

With no third VM there is one runner, as on the laptop today. The queue still takes every job and prices them in order. A second runner is a later decision and would be a second Windows VM without a card.

The Linux VM is Technical Design's.

Ubuntu LTS server under Hyper-V, administered over SSH by the Technical Design team. It carries the FastAPI services, Production Design Extraction and the Design Vault, and nothing that drives Excel, SOLIDWORKS or a SOLIDWORKS library; those it reaches on VM 1 over HTTPS. Ubuntu carries the Hyper-V integration drivers in its kernel, so nothing extra is installed.

Gotchas

None of these stops the plan. Each one is a setting or a habit, and each is on a list below with an owner.

1. **The card is VM 1's, and touching it means stopping VM 1.** A driver update inside the VM is an ordinary restart of the VM. Moving the card needs both VMs shut down. The host shows "Microsoft Basic Display" from then on; that is expected.
2. **One desktop in the Windows VM.** VM 1 is Windows 11 and allows one interactive session. A Remote Desktop into it as a different user pushes the runner off and can stall an estimate. Rule: connect as the runner account (same session) and leave by handing the session back to the console with `tscon`, or use the Hyper-V console from the host, which shows the runner's desktop without logging it off. The host's own two admin sessions are the normal way in.
3. **Everything on VM 1 shares one restart.** The runner, the portal, Client Briefing and Drawing Search go down together when VM 1 restarts, and a restart mid-run loses that estimate. Windows Update on VM 1 is scheduled out of hours and announced.
4. **Server drivers for a workstation board.** Scan supplies Windows 11 drivers for the 3XS. Windows Server 2025 installs on the Threadripper PRO board, but chipset and network drivers may need to come from the board maker. Ask Scan to confirm before the build week.
5. **SOLIDWORKS version and licence.** 2026 or later for Hyper-V 2025 support. A network licence is strongly preferred in a VM.
6. **The VMs must be on the office network, not behind the host.** An External virtual switch on the machine's network port, so both VMs get fixed addresses and DNS names, and browsers, Remote Desktop and SSH reach them by name.
7. **Sleep and lock in VM 1.** No sleep, no hibernate, no screen lock, no screensaver for the runner account, an exemption from any domain lock policy, and auto-logon with Sysinternals Autologon (Windows Hello off for that account).
8. **Linux under Hyper-V.** A Generation 2 Linux VM boots with Secure Boot only on the "Microsoft UEFI Certificate Authority" template; the Windows template refuses it. No desktop: SSH.
9. **GitLab on the Linux VM wants ports 80/443 and git over SSH.** Admin SSH and GitLab's SSH both want port 22; give GitLab its own port or its own address. PostgreSQL is opened only to VM 1 and the Linux VM itself, never to the office network.
10. **Licences add up.** Server 2025 Standard per core (two 16-core packs for this processor) and client access licences; Windows 11 Pro and Office for VM 1; nothing for the Linux VM. Server Standard's two free guest licences are for Windows *Server* guests, not Windows 11.

JACK

Infrastructure: the host and the two VMs

Everything below the applications: hardware, operating systems, network, accounts, backup. In build order.

THE HOST (WINDOWS SERVER 2025)

1. **Confirm with Scan** that Server 2025 drivers exist for the 3XS board, or where to get them.
2. **BIOS:** latest firmware; power on after AC loss; enable virtualisation (SVM) and IOMMU, which the GPU pass-through needs.
3. **Storage:** set up the two NVMe drives (mirror or split OS / VMs — Jack's call) and install Windows Server 2025 with current drivers.
4. **Network and domain:** fixed IP, DNS name, join the domain.
5. **Hyper-V:** add the Hyper-V role and create an **External virtual switch** on the machine's network port, so the VMs sit on the office network.
6. **SOLIDWORKS licence manager** (SolidNetWork): install it on the host with the reseller's serial, and open ports `25734` and `25735` to VM 1 (and to designers' PCs if they share the pool). It needs SOLIDWORKS 2025 SP3 or later on Server 2025, and must be at least as new as the SOLIDWORKS install.
7. **Admin access for James:** Remote Desktop to the host as an administrator (two sessions), and the host's admin share, which the laptop's `push-to-server.ps1` copies through.

VM 1 (WINDOWS 11 PRO, A VIRTUAL MACHINE ON THE SERVER HOST): THE GRAPHICS CARD AND EVERYTHING PEOPLE USE

8. **Create it as Generation 2** with Secure Boot and a virtual TPM (Windows 11 requires both).
9. **Fixed resources:** 16 virtual CPUs, 64 GB memory with Dynamic Memory **off**, a 500 GB+ virtual disk on NVMe; automatic start with the host, automatic stop action "turn off" (DDA requires it).
10. **Pass the RTX 4000 through** (Discrete Device Assignment): disable the card on the host, dismount it, assign it to the VM, then install NVIDIA's RTX workstation driver inside the VM. The host and the Linux VM then have no graphics card; that is expected.
11. **Windows 11 Pro**, fixed IP and DNS name on the External switch, joined to the domain, fully updated.
12. **Runner account:** a dedicated domain user that logs on automatically at boot (Sysinternals Autologon, Windows Hello off for it); read/write to the estimating shares and read to W: by UNC path (`\\server\share`), not a drive letter).
13. **Power and lock policy:** no sleep, no hibernate, no screen lock, no screensaver, exempt from any domain lock policy — SOLIDWORKS and Excel need a live desktop session.
14. **Firewall on VM 1:** allow the estimating portal (`8071`), Drawing Search (`5000`) and Client Briefing's port from the office network; allow the COM tools' port from the Linux VM only.
15. **Windows Update on VM 1:** restarts scheduled out of hours and announced to James.
16. A `tscon` **hand-back script** on the runner's desktop, and James shown the Hyper-V console route in.

VM 2 (LINUX): TECHNICAL DESIGN

17. **Create it as Generation 2** with Secure Boot on the **Microsoft UEFI Certificate Authority** template; no virtual TPM needed. 8 virtual CPUs, 32 GB memory, a 1 TB+ virtual disk on NVMe that can be grown; automatic start with the host.
18. **Ubuntu LTS server**, fixed IP and DNS name on the External switch, SSH open to the Technical Design team with sudo accounts, fully updated. **No graphics card.**
19. **Firewall on VM 2:** GitLab (80 / 443), and its git-over-SSH port) and the FastAPI services' port from the office network; PostgreSQL (5432) from VM 1 only.
20. **A read-only mount of W:** for the FastAPI services (a service account on the file server; the credentials kept on the VM, not in any script).

RUNNING IT

21. **Licences:** Windows Server 2025 Standard per core (two 16-core packs for the 9975WX) and client access licences; one Windows 11 Pro and one Office licence for VM 1; nothing for the Linux VM.
22. **Backups:** host and both VMs (Windows Server Backup or the company tool, with the VMs exported); the Design Vault's GitLab backup on its own schedule; one restore of each tested before go-live.
23. **Windows Update:** host and VM 1 on different nights, James told first. Ubuntu's security updates unattended, reboots announced.
24. **Antivirus:** apply the exclusions James supplies (SOLIDWORKS temp, the output folders, Drawing Search's index).
25. **Power:** on the UPS, set to start again after a power cut, both VMs starting automatically.

Handover checks

Jack confirms these, then the machines are James's:

- In VM 1's Device Manager the graphics card shows as **NVIDIA RTX 4000 Ada** with the NVIDIA driver, not "Microsoft Basic Display".
- After a full restart of the host, both VMs start by themselves; VM 1 logs in as the runner account with nobody touching it and is still unlocked an hour later.
- Signed in as the runner account, VM 1 can open and save a file in the estimating folders and open a file on W: by UNC path.
- From VM 1, the licence manager on the host answers on port 25734.
- From a desk, VM 1 answers on ports 8071 and 5000 (once James's services run, this is just a browser test); from the Linux VM, VM 1 answers on the COM tools' port.
- From James's laptop: Remote Desktop to the host as an administrator while another admin is already connected; the Hyper-V console shows VM 1's desktop; SSH into the Linux VM by name, and the W: mount lists a job folder.

Applications: the laptop premise

Everything that makes it an SDI Intelligence machine, installed the same way as on the laptop. The Technical Design team does the Linux side with James.

ON VM 1 (ESTIMATING, CLIENT ESTIMATING, CLIENT BRIEFING, DRAWING SEARCH)

1. **SOLIDWORKS:** install and point it at the host's licence manager. Open one model to confirm the graphics card is in use.
2. **Office / Excel:** install and activate; open one estimate book.
3. **PDM / Design Vault client,** if models come from the vault.
4. **Python and Git,** clone the repository, create the engine's `.venv`.
5. **The `.env` file,** written by hand for this machine — never copied from the laptop.
6. **The estimating portal:** the backend's own `.venv` and `.env`; install and start the SDIIntelligence service on `8071`; check it loads from a desk.
7. **The runner:** scheduled task at logon (`restart-runner.ps1`) pointing at the portal on this machine.
8. **Client Briefing Intelligence:** install as on its current machine, its own `.env` and port, started as a service; Jonathan's approval step tested end to end.
9. **Drawing Search:** install on port `5000`, point it at W: by UNC path, build the Fixture Library index on VM 1's own disk (455,000 fixtures: allow a day), confirm the image search is using the card.
10. **Technical Design's Windows parts:** the SOLIDWORKS Document Manager API (with its licence key from Dassault, under the terms confirmed below), the CAD Validator and the COM tools, installed beside SOLIDWORKS and exposed to the Linux VM over HTTPS on one port; test a DXF flat pattern from a model on W:.
11. **Hand Jack the antivirus exclusion paths** and the ports to open.

ON VM 2 (TECHNICAL DESIGN), WITH THE TECHNICAL DESIGN TEAM

12. **GitLab with LFS** on its own address or port, with backups scheduled; the SDI Design Vault's repositories created.
13. **PostgreSQL** for the Document Manager's intelligence layer, reachable from VM 1 only.
14. **The FastAPI services** (the Document Manager intelligence layer and Production Design Extraction) as systemd services, reading W: through the mount and calling VM 1's Document Manager API and COM tools over HTTPS.
15. **Prove one import:** a job from W: through the Document Manager into the vault, and one flat pattern requested from VM 1 and received.

PROVING THE WHOLE

16. **Run one known job** (12645 or 12173) on VM 1 and compare the book and the run time with the laptop's.
17. **Restart the host** and confirm both VMs, the portal, the runner, Client Briefing, Drawing Search and the Linux services all come back without anyone logging in.
18. **Remote in and out once** through the Hyper-V console while a short job runs on VM 1, and confirm the job finished.

19. **Retire the laptop's role:** the laptop becomes a development machine; the installed service and runner on it are stopped so there is one queue.

Agree together before the build week

Rule of thumb: if it would be the same on any Windows or Linux machine, it's Jack's. If it only exists because this is SDI Intelligence, it's James's.

QUESTION	WHO LEADS	WHY IT MATTERS
SOLIDWORKS licensing, in writing from the reseller before the build: (1) may this licence be activated and used in a Hyper-V VM; (2) does it permit automated, server-side use; (3) what licence is needed for unattended generation of STEP, PDF and DXF files. And, separately, the Document Manager API licence-key terms for SDI's use. Network (SolidNetWork) recommended, with the cost of converting	James, with the reseller	Hyper-V 2025 on Dassault's supported-platform list is technical support, not a licence right; the three rights are not inferred from it. A network licence is the only way to share SOLIDWORKS: seats in a pool on the host, first come, first served, shared with designers if wanted. A standalone licence is tied to one VM and can need reactivating when the VM changes. The runner holds a seat all day.
SOLIDWORKS version: 2026 or later	James, with the reseller	SOLIDWORKS lists Hyper-V 2025 (this host) for version 2026. The program runs only on Windows 10/11, never on the Server host.
Server 2025 drivers for the 3XS board	Jack, with Scan	Scan supplies Windows 11 drivers; the Server install may need the board maker's chipset and network drivers.
The Linux VM's disk: how much of W's history the Design Vault will hold	James, with Technical Design	Sets the virtual disk size and the backup size. It can be grown later, but the NVMe is 2 × 2 TB in total.
Ports and names: VM 1's portal, Client Briefing, Drawing Search and the HTTPS port for the Document Manager API and COM tools; the Linux VM's GitLab, FastAPI services and git-over-SSH.	James gives, Jack opens	Every link the estimators and the studio use names them.
Who is told before Windows updates or restarts, and the remote-access rule	Jack and James	A restart or a careless Remote Desktop session mid-run loses that estimate.

QUESTION	WHO LEADS	WHY IT MATTERS
A second graphics card for the Linux VM, and a second runner VM	James, later	Neither is in this build. A second card only if a model on the Linux side needs one; a second runner only if the queue backs up.